

Jennifer Liang

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Education

Harvard University

Cambridge, MA

A.B. IN APPLIED MATHEMATICS AND COMPUTER SCIENCE, GPA: 4.0/4.0

May 2023

- **Activities:** Women in Computer Science, Chinatown Citizenship, Harvard Crimson
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra and Real Analysis, Multivariable Calculus, Probability, Statistics, Functional Programming, Machine Learning, Social Computational Choice, Optimization, Stochastic Processes
- **Awards:** Phi Beta Kappa, Detur Book Prize, John Harvard Scholar

Work Experience

Two Sigma

New York, NY

QUANTITATIVE SOFTWARE ENGINEER

Aug. 2023 – Present

- Engineer on the portfolio construction team, building production systems for portfolio construction and market impact measurement.
- Designed and implemented workflows to report execution slippage, enhancing monitoring and model feedback loops across trading desk.
- Used Python-based analysis framework to perform quantitative diff testing and validation of a major market impact model release to macro.
- Integrated high-frequency market data features into optimization and accounting models and coordinated release to production. Delivered performance optimizations that reduced computational overhead from 50% to <5%, improving execution efficiency and unblocking PnL.
- Driving the development of new platform to streamline release of new predictive features for market impact modeling via Python-based models.

Two Sigma

New York, NY

SOFTWARE ENGINEER INTERN

May 2022 – Aug. 2022

- Built an interactive React web app for visualizing yield curves, pricing fixed income instruments, and analyzing key rate risk.
- Developed a parallelized REST API for yield curve construction and calculation of key rate risk, deployed to production.
- Gathered requirements from researchers and stakeholders to refine functionality, ensuring the tool addressed trading and research needs.

Bloomberg

Toronto, ON

SOFTWARE ENGINEER INTERN

May 2021 – July 2021

- Added UI enhancements to the Angular-based Enterprise Console, a platform for configuring and monitoring real-time financial data feeds.
- Extended Python backend services to automate validation of 80,000+ FIX connections via new API endpoints

Mobile Innovations

Toronto, ON

SOFTWARE ENGINEERING INTERN

May 2020 – Aug. 2020

- Implemented a courier delivery API using Node.js and SQL, enabling reliable order tracking and delivery management.
- Built out testing utilities in CURL, Node.js, JavaScript, and Python to automate endpoint validation.
- Migrated legacy web app to React, refactoring codebase and developing reusable components to improve maintainability and performance.

Leadership & Projects

Arctic Reading Circle

Toronto, ON

FOUNDER

Jul. 2020 – Jan. 2024

- Founded a Canadian non-profit delivering online programs to promote literacy and learning among newcomer youth.
- Organized read-alouds and workshops that helped students build confidence, communication, and leadership skills.
- Established outreach partnerships with organizations including the Afghan Women's Organization and TDSB Newcomer Services.

Harvard Women in CS

Cambridge, MA

CO-DIRECTOR, GIRLS WHO CODE

Apr. 2020 – Apr. 2022

- Led weekly coding workshops for 30+ students covering Scratch, Java, Python, and HTML.
- Created mentorship opportunities through the WiCS Pen Pal Program, connecting high school students with Harvard CS mentors.

Skills

Technical Java, Python, C/C++, JavaScript, TypeScript, Node, React, SQL, OCaml, Swift, Angular, Claude

Language Native English, Conversational Mandarin and French

Interests Running, Fencing, Kickboxing, Reading, Skiing, Canada Learning Code, Zombies Run

Olympiads European Girls' Math Olympiad, Asian Pacific Math Olympiads, Canadian Open Math Challenge